

Classification: Performance Metrics & Class Imbalance

Big Data y Machine Learning para Economía Aplicada

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Agenda

- 1 Recap
- 2 Other Algorithms for Probability Estimation
 - Non-Parametrics: K-Nearest Neighbors
 - Generative Models for Classification
 - Discriminant Analysis
 - Naive Bayes
- 3 Performance Metrics
 - Accuracy
 - TNR
 - TNR
- 4 ROC curve
- 5 Imbalanced Classification
 - Metrics

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Classification: Motivation

- ▶ Many predictive questions are about classification
 - ▶ Credit, Poverty, Firm default, Fraud, Unemployment, etc.
- ▶ Aim is to classify y , where y represents membership in a category
 - ▶ Qualitative, not necessarily ordered
 - ▶ We will focus for now in the binary case

*The prediction question is, given a new X ,
what is our best guess at the response category $\hat{y}$*

Classification: Recap

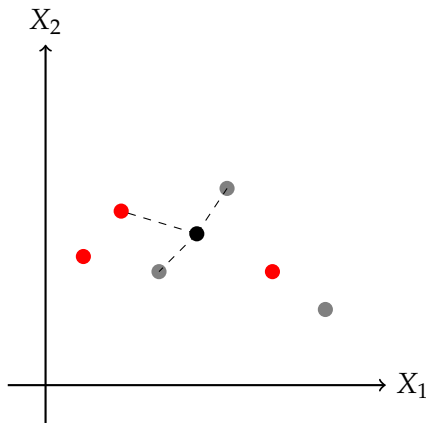
$$\hat{y}_i = 1[p_i \geq c]$$

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K-Nearest Neighbors

- K nearest neighbor (K-NN) algorithm predicts class $\hat{y}$ for x by asking *What is the most common class for observations around x ?*



K-Nearest Neighbors

- ▶ K nearest neighbor (K-NN) algorithm predicts class $\hat{y}$ for x by asking *What is the most common class for observations around x ?*
- ▶ Algorithm: given an input vector x_f where you would like to predict the class label
 - ▶ Find the K nearest neighbors in the dataset of labeled observations, $\{x_i, y_i\}_{i=1}^n$, the most common distance is the Euclidean distance:

$$d(x_i, x_f) = \sqrt{\sum_{j=1}^p (x_{ij} - x_{fj})^2} \quad (1)$$

- ▶ This yields a set of the K nearest observations with labels:

$$[x_{i1}, y_{i1}], \dots, [x_{iK}, y_{iK}] \quad (2)$$

- ▶ The predicted class of x_f is the most common class in this set

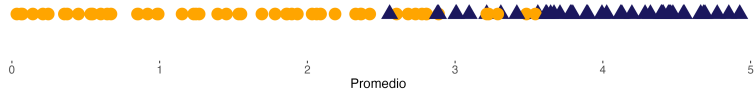
$$\hat{y}_f = \text{mode}\{y_{i1}, \dots, y_{iK}\} \quad (3)$$

Generative Models for Classification

- ▶ LDA
- ▶ QDA
- ▶ Naive Bayes

Discriminant Analysis

$$Pr(Y = 1|X) \quad (4)$$



Discriminant Analysis

The classification problem

- ▶ $Y \in \{0, 1\}$ the class label
- ▶ X a predictor (one to start)
- ▶ Our goal is to estimate the probability that an observation belongs to class 1, given its observed features:

$$\Pr(Y = 1 \mid X = x)$$

- ▶ is the **posterior class probability**.

Discriminant Analysis

Reverend Bayes to the rescue: Bayes Theorem

$$\Pr(Y = k \mid X = x) = \frac{\overbrace{\Pr(X = x \mid Y = k)}^{\text{class-conditional distribution}} \overbrace{\Pr(Y = k)}^{\text{class prior}}}{\underbrace{\Pr(X = x)}_{\text{marginal distribution of } X}}$$

- For classification, Bayes' theorem separates the problem into two quantities:
 - how frequent each class is

$$\pi_k = \Pr(Y = k)$$

- What values of X are typical within each class

$$\Pr(X = x \mid Y = k)$$

- Once these quantities are known, the posterior $\Pr(Y = k \mid X = x)$ follows from Bayes' theorem.

Discriminant Analysis

Bayes' theorem: continuous predictors

- Suppose that X is continuous, while

$$\Pr(X = x \mid Y = k) = 0,$$

- so we describe the distribution of X within each class using a **density**:

$$f_k(x) \equiv f_{X|Y}(x \mid Y = k).$$

- Bayes' theorem becomes

$$\Pr(Y = 1 \mid X = x) = \frac{f_1(x)\pi_1}{f_1(x)\pi_1 + f_0(x)\pi_0}$$

Discriminant Analysis

From Bayes' theorem to a classifier

To estimate

$$\Pr(Y = 1 \mid X = x),$$

we need to estimate

$$\pi_0, \pi_1$$

and the two class-conditional densities

$$f_0(x), f_1(x).$$

The class priors are straightforward:

$$\hat{\pi}_k = \frac{n_k}{n}.$$

Discriminant Analysis

From Bayes' theorem to a classifier

The harder problem is estimating

$$f_k(x) = f_{X|Y}(x \mid Y = k).$$

Discriminant Analysis

Linear Discriminant Analysis

LDA assumes that the predictors are Gaussian **within each class**.

For a single predictor,

$$X \mid Y = k \sim \mathcal{N}(\mu_k, \sigma^2)$$

so

$$f_k(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(x - \mu_k)^2}{2\sigma^2} \right].$$

Each class has its own mean,

$$\mu_0, \quad \mu_1,$$

but LDA assumes a **common variance**

$$\sigma^2$$

Example: LDA Unemployment



photo from <https://www.dailydot.com/parsec/batman-1966-labels-tumblr-twitter-vine/>

Linear Discriminant Analysis

Extensions

- ▶ If we have k predictors?
- ▶ then $X|Y \sim NM(\mu, \Sigma)$

$$f(X|Y = j) = \frac{1}{(2\pi)^{k/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu_j)' \Sigma_j (x - \mu_j)\right) \quad (5)$$

- ▶ μ_j is the vector of the sample means in each partition $j = 0, 1$
- ▶ Σ_j is the matrix of variance and covariances of each partition $j = 0, 1$

Linear Discriminant Analysis

- ▶ Why is it call linear?
- ▶ Note

$$p_i > \frac{1}{2} \iff \ln\left(\frac{p_i}{(1-p_i)}\right) > 0 \quad (6)$$

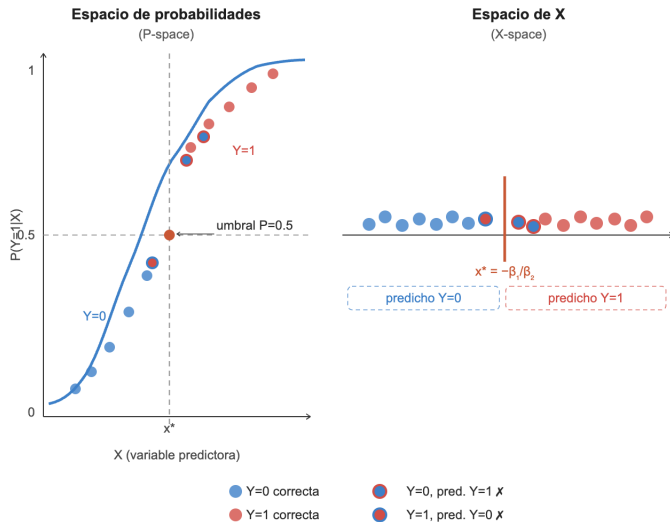
- ▶ Logit with one predictor

$$\beta_1 + \beta_2 X \quad (7)$$

- ▶ Classification: in the probability of space
- ▶ Discrimination: in the space of X
- ▶ $\beta_1 + \beta_2 X$ is the discrimination function for logit (it is lineal)

Linear Discriminant Analysis

Discrimination function



Linear Discriminant Analysis

- ▶ LDA?
- ▶ One predictor with $\sigma_0 = \sigma_1$ (equal variance)

$$Pr(Y = 1|X) = \frac{f(X|Y = 1)Pr(Y = 1)}{f(X|Y = 1)Pr(Y = 1) + f(X|Y = 0)(1 - Pr(Y = 1))} \quad (8)$$

- ▶ Then under the equal variance assumption

$$\frac{Pr(Y = 1|X)}{1 - Pr(Y = 1|X)} = \frac{f(X|Y = 1)Pr(Y = 1)}{f(X|Y = 0)(1 - Pr(Y = 1))} \quad (9)$$

$$= \frac{Pr(Y = 1)\exp(-(x - \mu_1)^2)}{(1 - Pr(Y = 1))\exp(-(x - \mu_0)^2)} \quad (10)$$

Linear Discriminant Analysis

► Taking logs

$$\log \left(\frac{Pr(Y = 1|X)}{1 - Pr(Y = 1|X)} \right) = \log \left(\frac{Pr(Y = 1)}{(1 - Pr(Y = 1))} + (x - \mu_1)^2 - (x - \mu_0)^2 \right) \quad (11)$$

$$= \log \left(\frac{Pr(Y = 1)}{(1 - Pr(Y = 1))} + \mu_1^2 - \mu_0^2 - 2(\mu_1 - \mu_0)x \right) \quad (12)$$

$$(13)$$

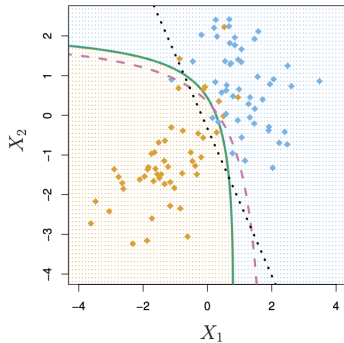
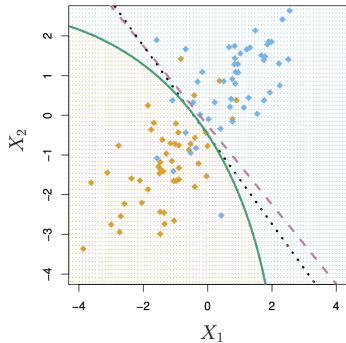
$$\log \left(\frac{Pr(Y = 1|X)}{1 - Pr(Y = 1|X)} \right) = \gamma_1 + \gamma_2 X \quad (14)$$

► under the assumption of equal variance the discrimination function is lineal

Discriminant Analysis

Quadratic Discriminant Analysis

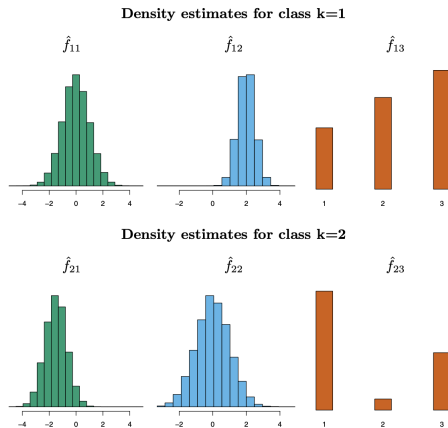
- QDA assumes different variances for the components



Naive Bayes

- NB assumes independence

$$f(X|Y = 1) = f(x_1|Y = 1) \times \cdots \times f(x_k|Y = 1) \quad (15)$$



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Confusion Matrix: Metrics

		y_i	
		1	0
$\hat{y}_i$	1	TP	FP
	0	FN	TN

Accuracy

		y_i	
		1	0
$\hat{y}_i$	1	TP	FP
	0	FN	TN

$$\frac{TP + TN}{TP + TN + FN + FP} \quad (16)$$

TNR

		y_i	
		1	0
$\hat{y}_i$	1	TP	FP
	0	FN	TN

$$P[\hat{y} = 0 | y = 0] = \frac{TN}{TN + FP} \quad (17)$$

TPR

		y_i	
		1	0
$\hat{y}_i$	1	TP	FP
	0	FN	TN

$$P[\hat{y} = 1 | y = 1] = \frac{TP}{TP + FN} \quad (18)$$

Example: Unemployment



photo from <https://www.dailydot.com/parsec/batman-1966-labels-tumblr-twitter-vine/>

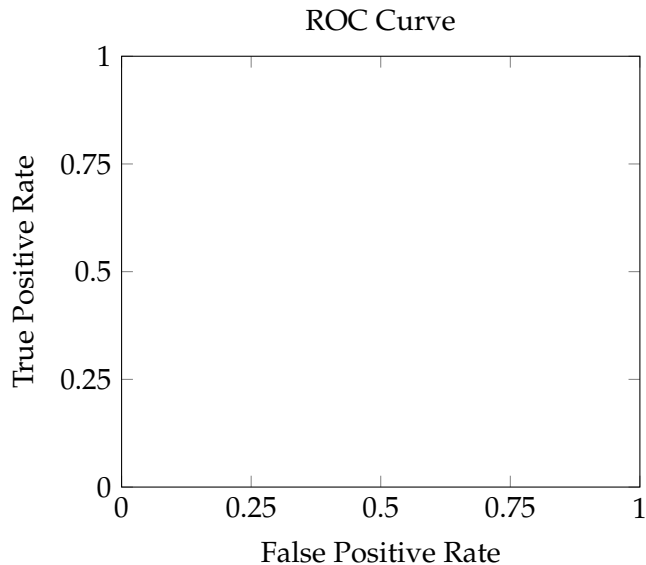
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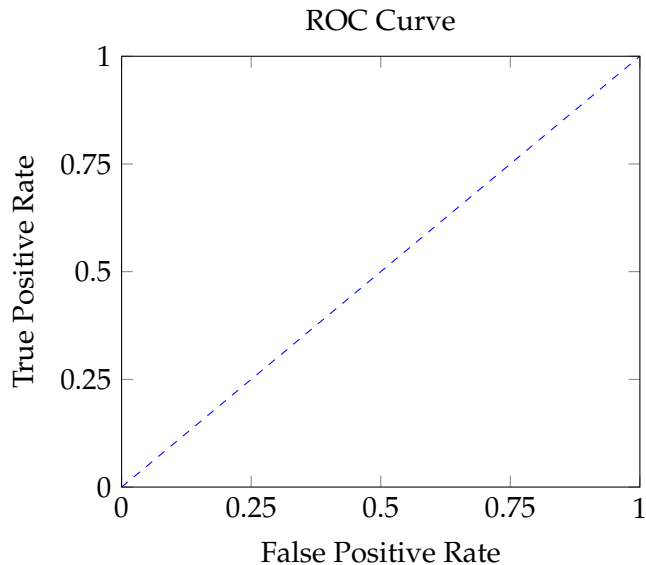
Trade-Off between Different Classification Thresholds

$$\hat{y}_i = 1[p_i \geq c]$$

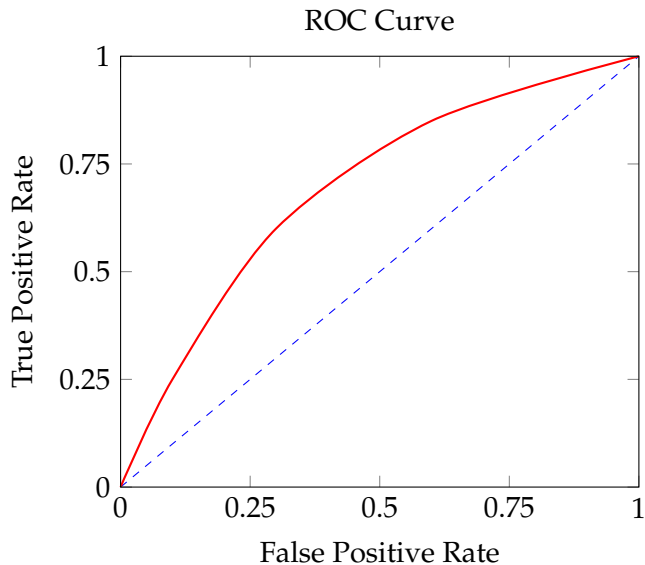
ROC Plot



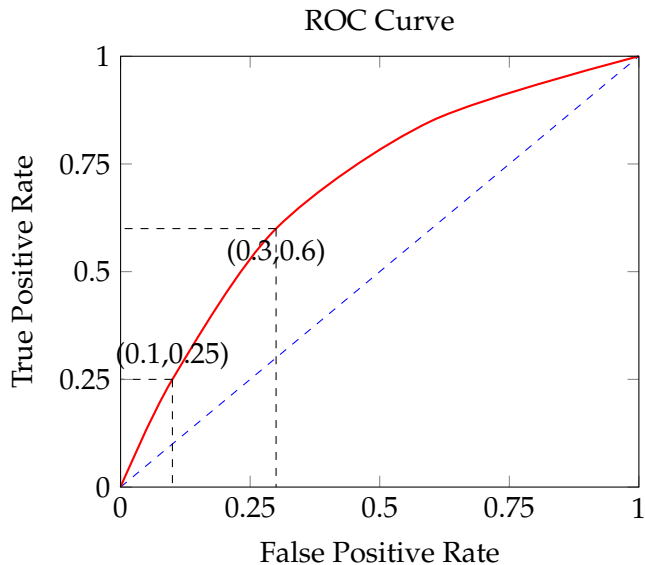
ROC Plot



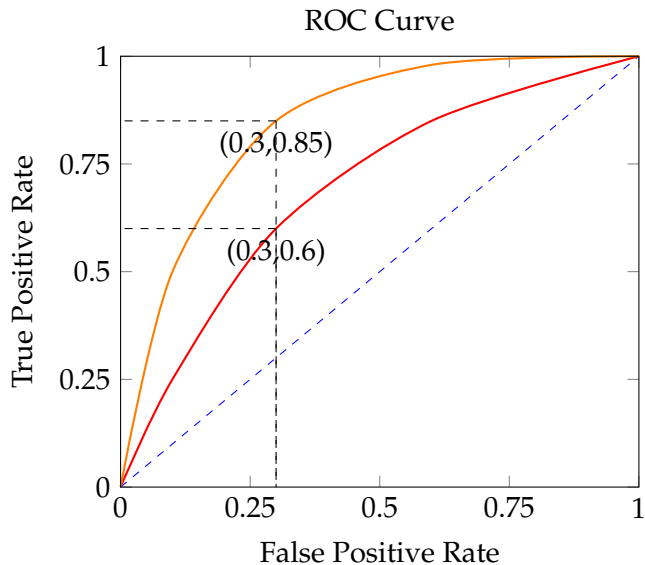
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ROC Plot



ROC Plot



Example: Unemployment



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Imbalanced Classification: Motivation

- ▶ Interest in one of the classes: Poor, Default, Unemployed, Fraud
- ▶ Imbalanced classes pose a challenge

Degree of imbalance	Proportion of Minority Class
Mild	20-40% of the data set
Moderate	1-20% of the data set
Extreme	<1% of the data set

TPR & PPV

		y_i	
		1	0
$\hat{y}_i$	1	TP	FP
	0	FN	TN

$$P[\hat{y} = 1 | y = 1] = \frac{TP}{TP + FN} \quad (19)$$

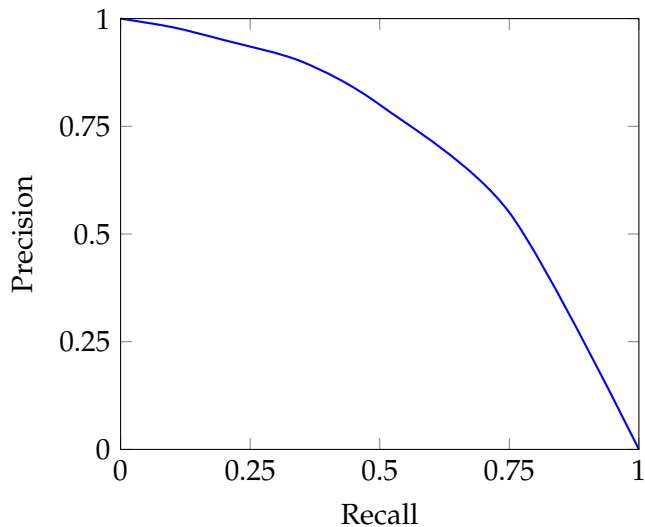
TPR & PPV

		y_i	
		1	0
$\hat{y}_i$	1	TP	FP
	0	FN	TN

$$P[\hat{y} = 1 | y = 1] = \frac{TP}{TP + FN} \quad (19)$$

$$P[y = 1 | \hat{y} = 1] = \frac{TP}{TP + FP} \quad (20)$$

PR-Curve



F-Scores

		y_i	
		1	0
$\hat{y}_i$	1	TP	FP
	0	FN	TN

$$F1 = 2 \frac{Precision \times Recall}{Precision + Recall} \quad (21)$$

F-Scores

		y_i	
		1	0
$\hat{y}_i$	1	TP	FP
	0	FN	TN

$$F_{\beta} = (1 + \beta^2) \frac{Precision \times Recall}{(\beta^2 \times Precision + Recall)} \quad (22)$$

Example: Unemployment



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